6.S892 Advanced Topics in Power Electronics Project Proposal

Extending DAB to multi-port DAB: effective design of integrated multi-winding transformers for decoupled power flow

Participant

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Background

The dual-active-bridge (DAB) converter, as equivalent circuit shown in Fig. 1, featuring a high-frequency (HF) transformer structure, together with the symmetrical H-bridges configuration on both the primary and secondary sides, provides a natural potential for extending the system to multiple ports by incorporating additional windings on the same transformer core. This leads to the development of multi-active-bridge (MAB) converters. The equivalent circuit of MAB can be found in Fig. 1. However, a key challenge in such multiport configurations lies in effectively decoupling the power flow among the different ports.

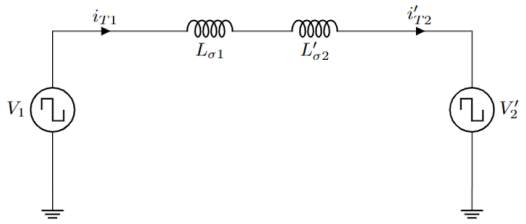


Fig 1: Equivalent model of a DAB converter, assuming significantly higher magnetizing inductance.

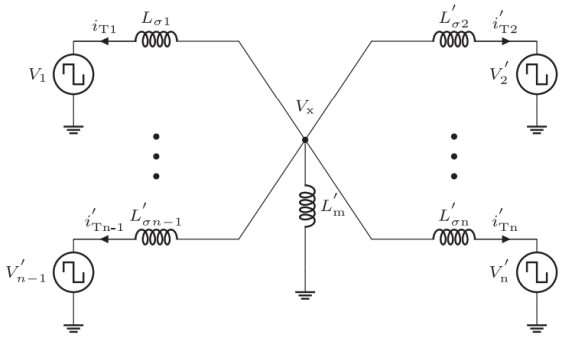


Fig 2: Equivalent model of an MAB converter [2].

One popular method is to add a series resonant capacitor to the primary side [1]. The other effective hardware approach to decoupling the ports is to introduce external inductors on selected sides, thereby creating a highly asymmetrical multi-winding structure, as shown in Fig. 3. This asymmetry allows one port to act as the "primary side," establishing the neutral point voltage, while the remaining ports function as "secondary sides." In this configuration, the power flow among ports becomes naturally decoupled.

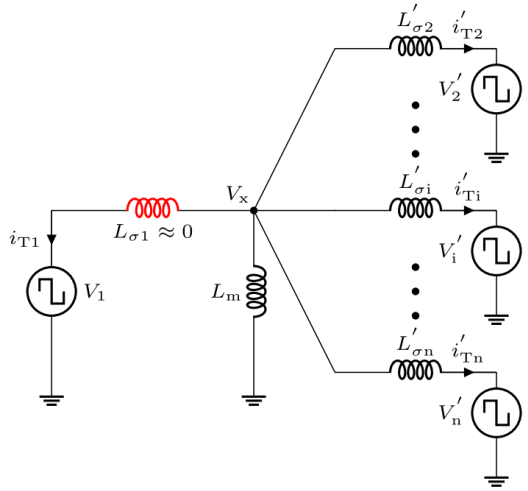


Fig 3: Adding external inductors on secondary sides to make the primary side leakage inductance negligible [2].

Both configurations mentioned above add additional components to the converter, which potentially increase the converter size and reduce the power density.

Research Target

Enlightened by some asymmetrical design examples for two-winding transformers in textbook, e.g. Fig. 3, Fig. 4, a single multi-winding integrated transformer design can achieve decoupled power flow by intentionally making primary side leakage significantly smaller than the secondary side leakage inductances, to achieve a very high asymmetrical transformer.

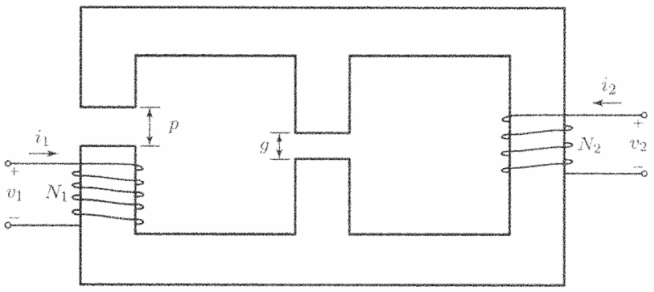


Fig. 3 Transformer example from KPVS Fig. 19.19 to make different leakage inductances between primary side and secondary side.

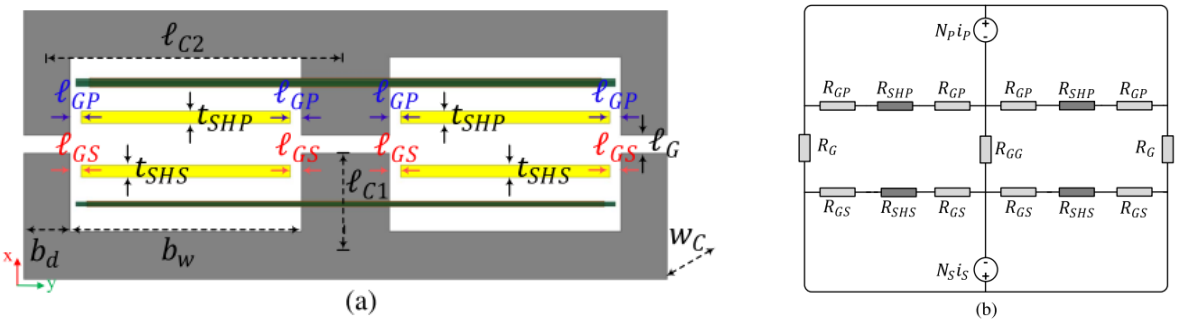


Fig. 4 Transformer design for asymmetrical leakage values [3].

Research Objectives

- ✓ Theoretically derive the optimal transformer.
- ✓ Perform FEM analysis to validate the proposed design.
- ✓ Verify system performance through simulations.

References

[1] P. Wang, X. Lu, W. Wang and D. Xu, "Hardware Decoupling and Autonomous Control of Series-Resonance-Based Three-Port Converters in DC Microgrids," in IEEE Transactions on Industry Applications, vol. 55, no. 4, pp. 3901-3914, July-Aug. 2019, doi: 10.1109/TIA.2019.2906112.

[2] S. Bandyopadhyay, P. Purgat, Z. Qin and P. Bauer, "A Multiactive Bridge Converter With Inherently Decoupled Power Flows," in IEEE Transactions on Power Electronics, vol. 36, no. 2, pp. 2231-2245, Feb. 2021, doi: 10.1109/TPEL.2020.3006266.

[3] S. A. Ansari, J. N. Davidson and M. P. Foster, "Fully-Integrated Transformer With Asymmetric Primary and Secondary Leakage Inductances for a Bidirectional Resonant Converter," in IEEE Transactions on Industry Applications, vol. 59, no. 3, pp. 3674-3685, May-June 2023, doi: 10.1109/TIA.2023.3252525.